

Medium_Articles ----- Medium - Articles Similarity.

Impoting Libraries

```
In [1]: import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np
import string

from IPython.display import display # libraries to display dataframe and images
from PIL import Image

sns.set_theme(style="darkgrid")
pd.set_option("display.max_columns", 100)
%matplotlib inline
```

```
In [44]: # Setting up the NLTK to pre-processing textual data

import nltk
from nltk.corpus import stopwords, wordnet
from nltk.stem import PorterStemmer, LancasterStemmer, SnowballStemmer, WordNetLemmm
from nltk.tokenize import word_tokenize, sent_tokenize
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger')
nltk.download('tagsets')
nltk.download('universal_tagset')
nltk.download('treebank')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\Home\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]   C:\Users\Home\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]   C:\Users\Home\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]   C:\Users\Home\AppData\Roaming\nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]   date!
[nltk_data] Downloading package tagsets to
[nltk_data]   C:\Users\Home\AppData\Roaming\nltk_data...
[nltk_data]   Package tagsets is already up-to-date!
[nltk_data] Downloading package universal_tagset to
[nltk_data]   C:\Users\Home\AppData\Roaming\nltk_data...
[nltk_data]   Package universal_tagset is already up-to-date!
[nltk_data] Downloading package treebank to
[nltk_data]   C:\Users\Home\AppData\Roaming\nltk_data...
[nltk_data]   Package treebank is already up-to-date!
```

Out[44]: True

```
In [3]: sns.set_theme(style="darkgrid")
pd.set_option("display.max_columns", 100)
%matplotlib inline
```

```
In [4]: ! pip install spacy
```

Requirement already satisfied: spacy in c:\users\home\anaconda3\lib\site-packages (3.8.14)

Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in c:\users\home\anaconda3\lib\site-packages (from spacy) (3.0.12)

Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in c:\users\home\anaconda3\lib\site-packages (from spacy) (1.0.5)

Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in c:\users\home\anaconda3\lib\site-packages (from spacy) (1.0.15)

Requirement already satisfied: cymem<2.1.0,>=2.0.2 in c:\users\home\anaconda3\lib\site-packages (from spacy) (2.0.13)

Requirement already satisfied: preshed<3.1.0,>=3.0.2 in c:\users\home\anaconda3\lib\site-packages (from spacy) (3.0.13)

Requirement already satisfied: thinc<8.4.0,>=8.3.12 in c:\users\home\anaconda3\lib\site-packages (from spacy) (8.3.13)

Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in c:\users\home\anaconda3\lib\site-packages (from spacy) (1.1.3)

Requirement already satisfied: srsly<3.0.0,>=2.5.3 in c:\users\home\anaconda3\lib\site-packages (from spacy) (2.5.3)

Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in c:\users\home\anaconda3\lib\site-packages (from spacy) (2.0.10)

Requirement already satisfied: weasel<2.0.0,>=1.0.0 in c:\users\home\anaconda3\lib\site-packages (from spacy) (1.0.0)

Requirement already satisfied: confection<2.0.0,>=1.3.2 in c:\users\home\anaconda3\lib\site-packages (from spacy) (1.3.3)

Requirement already satisfied: typer<1.0.0,>=0.3.0 in c:\users\home\anaconda3\lib\site-packages (from spacy) (0.20.0)

Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in c:\users\home\anaconda3\lib\site-packages (from spacy) (4.67.1)

Requirement already satisfied: numpy>=1.19.0 in c:\users\home\anaconda3\lib\site-packages (from spacy) (2.3.5)

Requirement already satisfied: requests<3.0.0,>=2.13.0 in c:\users\home\anaconda3\lib\site-packages (from spacy) (2.32.5)

Requirement already satisfied: pydantic<3.0.0,>=2.0.0 in c:\users\home\anaconda3\lib\site-packages (from spacy) (2.12.4)

Requirement already satisfied: jinja2 in c:\users\home\anaconda3\lib\site-packages (from spacy) (3.1.6)

Requirement already satisfied: setuptools in c:\users\home\anaconda3\lib\site-packages (from spacy) (80.9.0)

Requirement already satisfied: packaging>=20.0 in c:\users\home\anaconda3\lib\site-packages (from spacy) (25.0)

Requirement already satisfied: annotated-types>=0.6.0 in c:\users\home\anaconda3\lib\site-packages (from pydantic<3.0.0,>=2.0.0->spacy) (0.6.0)

Requirement already satisfied: pydantic-core==2.41.5 in c:\users\home\anaconda3\lib\site-packages (from pydantic<3.0.0,>=2.0.0->spacy) (2.41.5)

Requirement already satisfied: typing-extensions>=4.14.1 in c:\users\home\anaconda3\lib\site-packages (from pydantic<3.0.0,>=2.0.0->spacy) (4.15.0)

Requirement already satisfied: typing-inspection>=0.4.2 in c:\users\home\anaconda3\lib\site-packages (from pydantic<3.0.0,>=2.0.0->spacy) (0.4.2)

Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\home\anaconda3\lib\site-packages (from requests<3.0.0,>=2.13.0->spacy) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in c:\users\home\anaconda3\lib\site-packages (from requests<3.0.0,>=2.13.0->spacy) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\home\anaconda3\lib\site-packages (from requests<3.0.0,>=2.13.0->spacy) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in c:\users\home\anaconda3\lib\site-packages (from requests<3.0.0,>=2.13.0->spacy) (2025.11.12)

Requirement already satisfied: blis<1.4.0,>=1.3.0 in c:\users\home\anaconda3\lib\site-packages (from thinc<8.4.0,>=8.3.12->spacy) (1.3.3)

Requirement already satisfied: colorama in c:\users\home\anaconda3\lib\site-packages (from tqdm<5.0.0,>=4.38.0->spacy) (0.4.6)

Requirement already satisfied: click>=8.0.0 in c:\users\home\anaconda3\lib\site-packages (from typer<1.0.0,>=0.3.0->spacy) (8.2.1)

Requirement already satisfied: shellingham>=1.3.0 in c:\users\home\anaconda3\lib\site-packages (from typer<1.0.0,>=0.3.0->spacy) (1.5.4)

Requirement already satisfied: rich>=10.11.0 in c:\users\home\anaconda3\lib\site-packages (from typer<1.0.0,>=0.3.0->spacy) (14.2.0)

Requirement already satisfied: cloudpathlib>=0.7.0 in c:\users\home\anaconda3\lib\site-packages (from weasel<2.0.0,>=1.0.0->spacy) (0.24.0)

Requirement already satisfied: smart-open>=5.2.1 in c:\users\home\anaconda3\lib\site-packages (from weasel<2.0.0,>=1.0.0->spacy) (7.6.1)

Requirement already satisfied: httpx>=0.24.0 in c:\users\home\anaconda3\lib\site-packages (from weasel<2.0.0,>=1.0.0->spacy) (0.28.1)

Requirement already satisfied: anyio in c:\users\home\anaconda3\lib\site-packages (from httpx>=0.24.0->weasel<2.0.0,>=1.0.0->spacy) (4.10.0)

Requirement already satisfied: httpcore==1.* in c:\users\home\anaconda3\lib\site-packages (from httpx>=0.24.0->weasel<2.0.0,>=1.0.0->spacy) (1.0.9)

Requirement already satisfied: h11>=0.16 in c:\users\home\anaconda3\lib\site-packages (from httpcore==1.*->httpx>=0.24.0->weasel<2.0.0,>=1.0.0->spacy) (0.16.0)

Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\home\anaconda3\lib\site-packages (from rich>=10.11.0->typer<1.0.0,>=0.3.0->spacy) (2.2.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\home\anaconda3\lib\site-packages (from rich>=10.11.0->typer<1.0.0,>=0.3.0->spacy) (2.19.2)

Requirement already satisfied: mdurl~=0.1 in c:\users\home\anaconda3\lib\site-packages (from markdown-it-py>=2.2.0->rich>=10.11.0->typer<1.0.0,>=0.3.0->spacy) (0.1.2)

Requirement already satisfied: wrapt in c:\users\home\anaconda3\lib\site-packages (from smart-open>=5.2.1->weasel<2.0.0,>=1.0.0->spacy) (1.17.0)

Requirement already satisfied: sniffio>=1.1 in c:\users\home\anaconda3\lib\site-packages (from anyio->httpx>=0.24.0->weasel<2.0.0,>=1.0.0->spacy) (1.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in c:\users\home\anaconda3\lib\site-packages (from jinja2->spacy) (3.0.2)

```
In [5]: import spacy # Tokenization, Part-of-Speech (POS) tagging, Named Entity Recognition
        from spacy import displacy
```

Uploading the dataset

```
In [6]: articles = pd.read_csv("medium_articles_v3.csv")
        display(articles.head(10))
        print("Shape of dataframe : {}".format(articles.shape))
```

	link	title	sub_title	author	rea
0	https://towardsdatascience.com/ensemble-method...	Ensemble methods: bagging, boosting and stacking	Understanding the key concepts of ensemble lea...	Joseph Rocca	
1	https://towardsdatascience.com/understanding-a...	Understanding AUC - ROC Curve	In Machine Learning, performance measurement i...	Sarang Narkhede	
2	https://towardsdatascience.com/how-to-work-wit...	How to work with object detection datasets in ...	A comprehensive guide to defining, loading, ex...	Eric Hofesmann	
3	https://towardsdatascience.com/11-dimensional...	11 Dimensionality reduction techniques you sho...	Reduce the size of your dataset while keeping ...	Rukshan Pramoditha	
4	https://towardsdatascience.com/the-time-series...	The Time Series Transformer	Attention Is All You Need they said. Is it a m...	Theodoros Ntakouris	
5	https://netflixtechblog.com/learning-a-persona...	Learning a Personalized Homepage	how to best tailor each member's homepage to m...	Netflix Technology Blog	
6	https://towardsdatascience.com/6-data-science-...	6 Data Science Certificates To Level Up Your C...	Pump up your portfolio and get closer to your ...	Sara A. Metwalli	
7	https://towardsdatascience.com/transformers-ex...	Transformers Explained Visually (Part 2): How ...	A Gentle Guide to the Transformer under the ho...	Ketan Doshi	
8	https://medium.com/coders-camp/60-python-proje...	60 Python Projects with Source Code	60 Python Projects with Source code solved and...	Aman Kharwal	
9	https://towardsdatascience.com/geometric-found...	Geometric foundations of Deep Learning	Geometric Deep Learning is an attempt to unify...	Michael Bronstein	

link	title	sub_title	author	re

Shape of dataframe : (208, 7)

In []:

```
In [7]: from pprint import pprint
        pprint(articles.iloc[1].to_dict(), compact=True)
```

```
{'author': 'Sarang Narkhede',
'id': 2,
'link': 'https://towardsdatascience.com/understanding-auc-roc-curve-68b2303cc9c5?source=tag_archive-----1-----',
'reading_time': 5,
'sub_title': 'In Machine Learning, performance measurement is an essential '
'task. So when it comes to a classification problem, we can '
'count on an AUC - ROC Curve. When we need to check or visualize '
'the performance...',
'text': 'In Machine Learning, performance measurement is an essential task. '
'So when it comes to a classification problem, we can count on an AUC '
'- ROC Curve. When we need to check or visualize the performance of '
'the multi-class classification problem, we use the AUC (Area Under '
'The Curve) ROC (Receiver Operating Characteristics) curve. It is one '
'of the most important evaluation metrics for checking any '
"classification model's performance. It is also written as AUROC "
'(Area Under the Receiver Operating Characteristics)\n'
'Note: For better understanding, I suggest you read my article about '
'Confusion Matrix.\n'
'This blog aims to answer the following questions:\n'
'1. What is the AUC - ROC Curve?\n'
'2. Defining terms used in AUC and ROC Curve.\n'
'3. How to speculate the performance of the model?\n'
'4. Relation between Sensitivity, Specificity, FPR, and Threshold.\n'
'5. How to use AUC - ROC curve for the multiclass model?\n'
'AUC - ROC curve is a performance measurement for the classification '
'problems at various threshold settings. ROC is a probability curve '
'and AUC represents the degree or measure of separability. It tells '
'how much the model is capable of distinguishing between classes. '
'Higher the AUC, the better the model is at predicting 0 classes as 0 '
'and 1 classes as 1. By analogy, the Higher the AUC, the better the '
'model is at distinguishing between patients with the disease and no '
'disease.\n'
'The ROC curve is plotted with TPR against the FPR where TPR is on '
'the y-axis and FPR is on the x-axis.\n'
'An excellent model has AUC near to the 1 which means it has a good '
'measure of separability. A poor model has an AUC near 0 which means '
'it has the worst measure of separability. In fact, it means it is '
'reciprocating the result. It is predicting 0s as 1s and 1s as 0s. '
'And when AUC is 0.5, it means the model has no class separation '
'capacity whatsoever.\n'
"Let's interpret the above statements.\n"
"As we know, ROC is a curve of probability. So let's plot the "
'distributions of those probabilities:\n'
'Note: Red distribution curve is of the positive class (patients with '
'disease) and the green distribution curve is of the negative '
'class(patients with no disease).\n'
"This is an ideal situation. When two curves don't overlap at all "
'means model has an ideal measure of separability. It is perfectly '
'able to distinguish between positive class and negative class.\n'
'When two distributions overlap, we introduce type 1 and type 2 '
'errors. Depending upon the threshold, we can minimize or maximize '
'them. When AUC is 0.7, it means there is a 70% chance that the model '
'will be able to distinguish between positive class and negative '
'class.\n'
'This is the worst situation. When AUC is approximately 0.5, the '
```

```
'model has no discrimination capacity to distinguish between positive '
'class and negative class.\n'
'When AUC is approximately 0, the model is actually reciprocating the '
'classes. It means the model is predicting a negative class as a '
'positive class and vice versa.\n'
'Sensitivity and Specificity are inversely proportional to each '
'other. So when we increase Sensitivity, Specificity decreases, and '
'vice versa.\n'
'Sensitivity, Specificity and Sensitivity, Specificity\n'
'When we decrease the threshold, we get more positive values thus it '
'increases the sensitivity and decreasing the specificity.\n'
'Similarly, when we increase the threshold, we get more negative '
'values thus we get higher specificity and lower sensitivity.\n'
'As we know FPR is 1 - specificity. So when we increase TPR, FPR also '
'increases and vice versa.\n'
'TPR, FPR and TPR, FPR\n'
'In a multi-class model, we can plot the N number of AUC ROC Curves '
'for N number classes using the One vs ALL methodology. So for '
'example, If you have three classes named X, Y, and Z, you will have '
'one ROC for X classified against Y and Z, another ROC for Y '
'classified against X and Z, and the third one of Z classified '
'against Y and X.\n'
'Thanks for Reading.\n'
'I hope I've given you some understanding of what exactly is the AUC "
'- ROC Curve. If you like this post, a tad of extra motivation will '
'be helpful by giving this post some claps . I am always open to your '
'questions and suggestions. You can share this on Facebook, Twitter, '
'Linkedin, so someone in need might stumble upon this.\n'
'You can reach me at:\n'
'LinkedIn: https://www.linkedin.com/in/narkhedesarang/\n'
'Twitter: https://twitter.com/narkhede_sarang\n'
'Github: https://github.com/TheSarang\n',
'title': 'Understanding AUC - ROC Curve'}
```

In []:

Statistical summary

In [8]: `articles.describe(include='all')`

Out[8]:

	link	title	sub_title	author	reading_time
count	208	208	208	208	208.000000
unique	208	208	204	179	NaN
top	https://towardsdatascience.com/ensemble-method...	Ensemble methods: bagging, boosting and stacking	Update: This article is part of a series. Chec...	Adam Geitgey	NaN
freq	1	1	4	5	NaN
mean	NaN	NaN	NaN	NaN	12.375000
std	NaN	NaN	NaN	NaN	13.880224
min	NaN	NaN	NaN	NaN	2.000000
25%	NaN	NaN	NaN	NaN	6.000000
50%	NaN	NaN	NaN	NaN	9.000000
75%	NaN	NaN	NaN	NaN	13.000000
max	NaN	NaN	NaN	NaN	149.000000

In []:

Ditsribution of reading times(only integer variable)

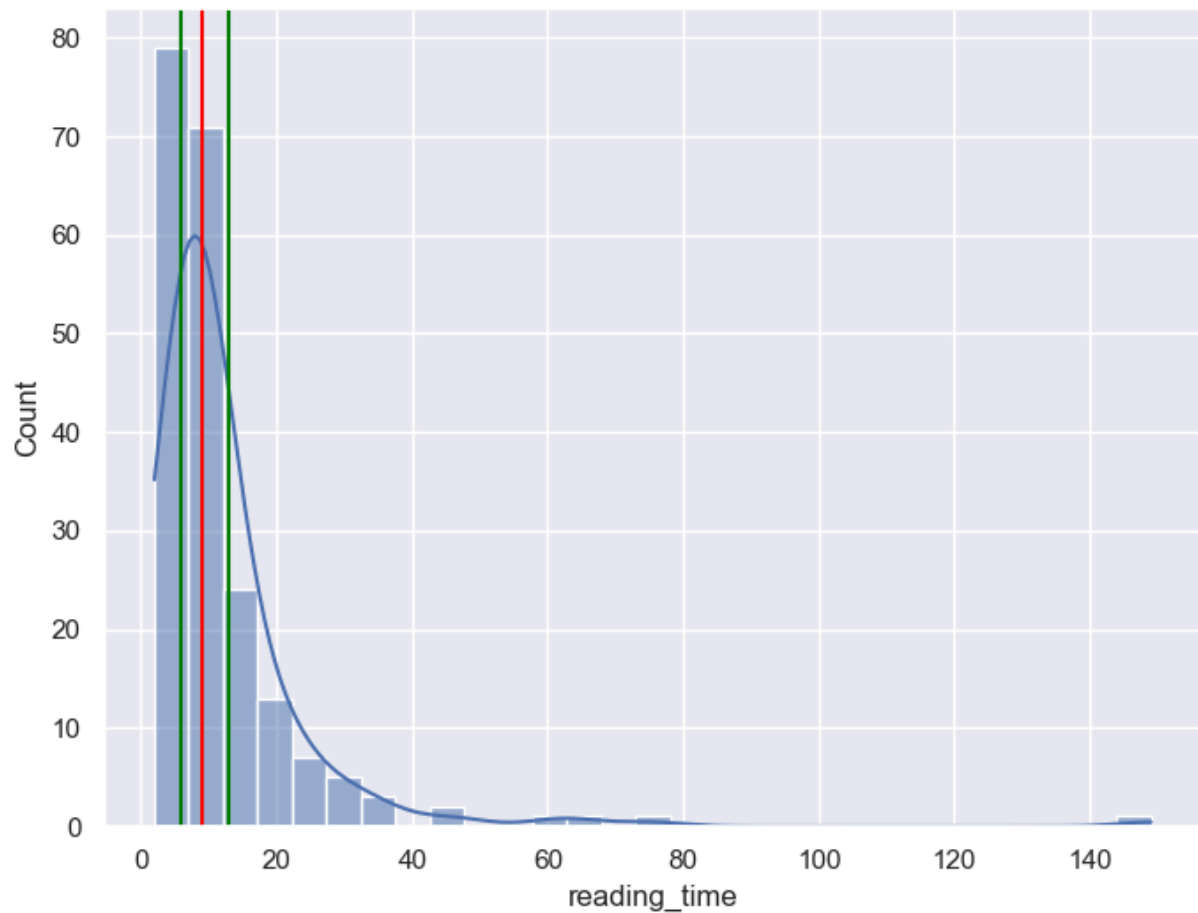
```
In [9]: # ditsribution of reading times in our corpus

fig, axes = plt.subplots(figsize = (8, 6))
sns.histplot(articles["reading_time"], kde=True, ax = axes)

# Computing percentile of the reading_time data.

first_q = np.percentile(articles["reading_time"], 25)
second_q = np.percentile(articles["reading_time"], 50)
third_q = np.percentile(articles["reading_time"], 75)
plt.axvline(first_q, color = "green")
plt.axvline(second_q, color = "red")
plt.axvline(third_q, color = "green")
plt.suptitle("Distribution of reading time across articles")
plt.show()
```

Distribution of reading time across articles



It is a right skewed distribution, indicating that the majority of articles are quick read, while a few are very long. The median reading time is 9 minutes

In []:

Relating the reading time with article length

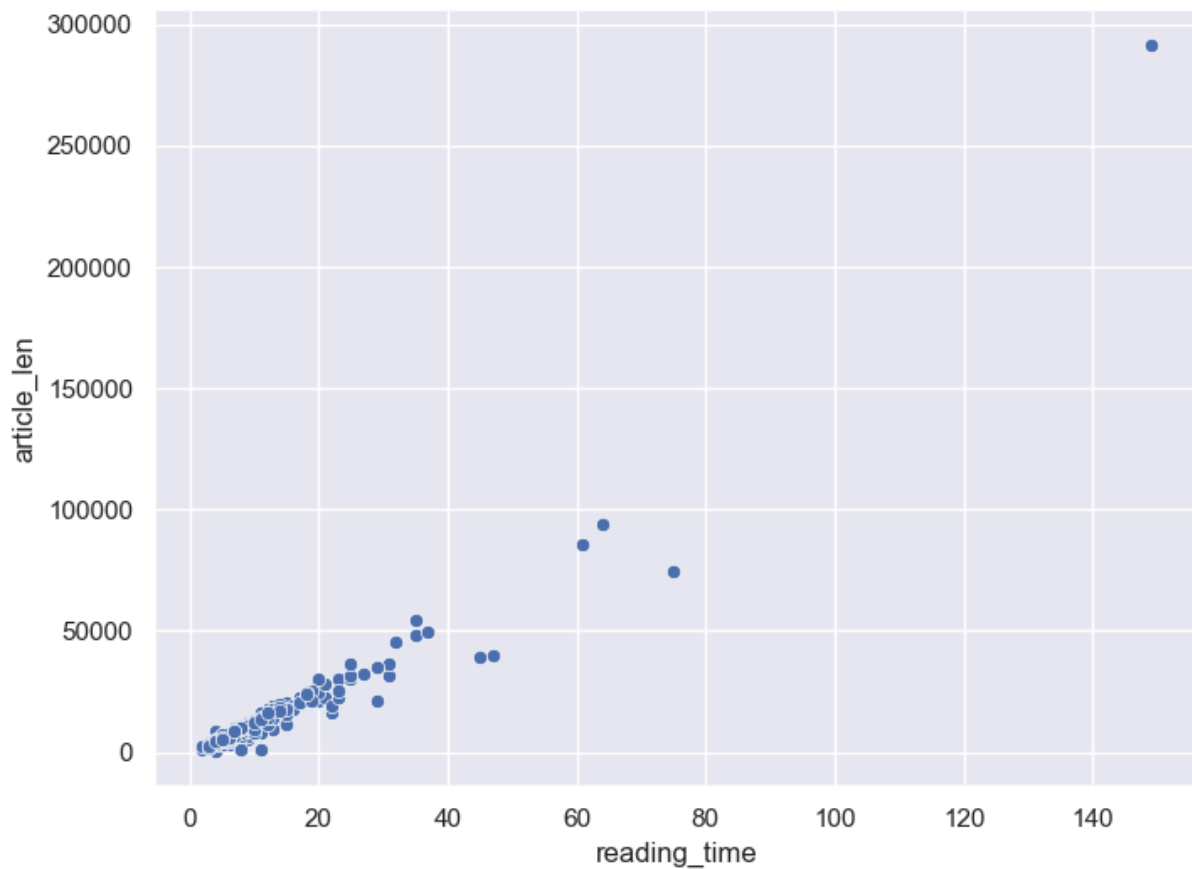
```
In [10]: from scipy.stats import pearsonr

articles["article_len"] = articles["text"].apply(lambda x : len(x))

# scatterplot
fig, axes = plt.subplots(figsize = (8, 6))
sns.scatterplot(x = articles["reading_time"], y = articles["article_len"])
# Pearson correlation coefficient to measure the linear relationship between two se
corr_coeff, _ = pearsonr(articles["reading_time"], articles["article_len"])

plt.suptitle("Reading Time vs. Article Length | Corr. Coeff : {}".format(round(corr_coeff, 2)))
plt.show()
```

Reading Time vs. Article Length | Corr. Coeff : 0.96



The reading time and article length are highly correlated.

In []:

similarity between articles on similar topics

```
In [11]: for author, count in dict(articles["author"].value_counts()).items():
          if(count < 2):
              continue
          print("Articles by {} :".format(author))
          for title in articles[articles["author"] == author]["title"].values:
              print(title)
          print("-"*120)
```

Articles by Adam Geitgey :

Machine Learning is Fun Part 5: Language Translation with Deep Learning and the Magic of Sequences

Machine Learning is Fun! Part 4: Modern Face Recognition with Deep Learning

Machine Learning is Fun! Part 3: Deep Learning and Convolutional Neural Networks

Machine Learning is Fun! Part 2

Machine Learning is Fun Part 6: How to do Speech Recognition with Deep Learning

Articles by Joseph Rocca :

Ensemble methods: bagging, boosting and stacking

Understanding Variational Autoencoders (VAEs)

Understanding Generative Adversarial Networks (GANs)

Articles by Natassha Selvaraj :

I tripled my income with data science. Here's how.

How to Land a Data Analytics Job in 6 Months

Top 10 Data Science Projects for Beginners

Articles by Chris Zaire :

5 Online Courses I Took as a Self-Taught Data Scientist

Is Data Science Still a Rising Career in 2021

Articles by Dario Radecic :

Are The New M1 Macbooks Any Good for Data Science? Let's Find Out

Top 3 Reasons Why I Sold My M1 Macbook Pro as a Data Scientist

Articles by Adi Bronshtein :

Simple and Multiple Linear Regression in Python

Train/Test Split and Cross Validation in Python

Articles by Serge Faguet :

How to biohack your intelligence with everything from sex to modafinil to MDMA

I'm 32 and spent \$200k on biohacking. Became calmer, thinner, extroverted, healthier & happier.

Articles by Will Koehrsen :

Hyperparameter Tuning the Random Forest in Python

Random Forest in Python

Articles by Madison Hunter :

15 Habits I Learned from Highly Effective Data Scientists

How to Study for the Google Data Analytics Professional Certificate

Articles by Andrei Tapalaga :

Chernobyl's Blown Up Reactor 4 Just Woke Up

Eben Byers: The Man Who Drank Radioactive Water Until His Jaw Fell Off

Articles by Susan Li :

Building A Logistic Regression in Python, Step by Step

An End-to-End Project on Time Series Analysis and Forecasting with Python

Articles by Yueh :

Richart + @GoGo

2018 & : KOKO COMBO icash

Articles by Daniel Jeffries :

How to Crush the Crypto Market, Quit Your Job, Move to Paradise and Do Whatever You
Want the Rest of Your Life

Why People Still Don't Get Cryptocurrency

Articles by allen farrington :

Bitcoin Is Venice

Gauge Theory Does Not Fix This

Articles by Alexandr Nellson :

How to invest in Bitcoin properly. Blockchain and other cryptocurrencies

How to store Bitcoins and other cryptocurrencies properly.

Articles by Mercatus Center :

Nassim Nicholas Taleb on Self-Education and Doing the Math (Ep. 41 Live at Mercatu
s)

Cliff Asness on Marvel vs. DC and Why Never to Share a Gym with Cirque du Soleil (E
p. 5 Live at Mason)

Articles by Blake Ross :

Wealthfront:Silicon Valley Tech at Wall Street Prices

Heroes Give, Superheroes Borrow

Articles by Tomas Pueyo :

Coronavirus: Why You Must Act Now

Coronavirus: The Hammer and the Dance

Articles by Terence Shin :

OVER 100 Data Scientist Interview Questions and Answers!

50+ Statistics Interview Questions and Answers for Data Scientists for 2022

Articles by Rukshan Pramoditha :

11 Dimensionality reduction techniques you should know in 2021

20 Necessary Requirements of a Perfect Laptop for Data Science and Machine Learning
Tasks

Articles by Ketan Doshi :

Transformers Explained Visually (Part 2): How it works, step-by-step
 Transformers Explained Visually (Part 3): Multi-head Attention, deep dive

Articles by Aman Kharwal :
 60 Python Projects with Source Code
 180 Data Science and Machine Learning Projects with Python

Articles by Mahmoud Harmouch :
 17 Clustering Algorithms Used In Data Science and Mining
 17 types of similarity and dissimilarity measures used in data science.

Articles by Sara A. Metwalli :
 6 Data Science Certificates To Level Up Your Career
 6 Machine Learning Certificates to Pursue in 2021

From the above output, we can make out that not only the authors tend to write multiple articles on the same topic, but also, there are multiple authors writing on the same topic.

In []:

Preprocessing

In [16]: `!pip install contractions -q`

In [25]: `import contractions
import re`

In [26]: `def process_sentence(sentence, nlp_object):
 # Convert to Lowercase
 sentence = sentence.lower()

 # Expanding contractions
 sentence = contractions.fix(sentence)

 # Lemmatization and removing stopwords
 doc = nlp_object(sentence)
 sentence = " ".join([token.lemma_ for token in doc if not token.is_stop])

 # Remove punctuation
 for p in string.punctuation:
 sentence = sentence.replace(p, " ")
 sentence = re.sub(r"\s+", " ", sentence) # Replace all whitespace characters with a single space

 return sentence`

In [27]: `# tqdm to see real time progress

from tqdm.notebook import tqdm
tqdm.pandas()`

```

nlp = spacy.load('en_core_web_sm') # English pipeline optimized for CPU

def process_article(article_text, nlp_object):
    processed_article_sentences = []
    # using nltk sentence tokenizer
    for sentence in sent_tokenize(article_text):
        # preprocessing each sentence using our process_sentence function
        processed_article_sentences.append(process_sentence(sentence, nlp_object))
    # joining preprocessed sentence as a complete paragraphs of the article
    return " ".join(processed_article_sentences)

articles["processed_text"] = articles["text"].progress_apply(lambda x : process_art
0%|          | 0/208 [00:00<?, ?it/s]

```

Preprocessing and tokenization of the article text done.

In []:

Similarity between documents

Similarity using BOW (Bag of Words)

```

In [33]: # BOW representation of the dataset Using CountVectorizer from scikit-learn

from sklearn.feature_extraction.text import CountVectorizer
count_vectorizer = CountVectorizer(min_df=5)

# min_df: ignore terms that have a document frequency strictly lower than the given

# Learn the vocabulary dictionary and return document-term matrix
bow_features = count_vectorizer.fit_transform(articles["processed_text"]).todense()

# create dataframe
bow_features_df = pd.DataFrame(bow_features)
bow_features_df.columns = count_vectorizer.get_feature_names_out() # Get output fea
bow_features_df["TITLE"] = articles["title"]
bow_features_df["ID"] = articles["id"]
display(bow_features_df)

```

	00	000	01	05	06	07	10	100	1000	101	10k	10x	11	12	120	125	128	13
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
203	1	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0
204	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
205	0	0	0	0	0	0	1	0	0	0	0	0	1	1	0	0	0	0
206	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0
207	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0

208 rows × 3715 columns



Visualize this high dimensional data

```
In [34]: from sklearn.manifold import TSNE

# using t-sne to observe any trends, and clusters.
tsne = TSNE(n_components=2) # n_components: estimated number of components
tsne_bow_features = tsne.fit_transform(bow_features_df[count_vectorizer.get_feature
tsne_bow_features_df = pd.DataFrame(tsne_bow_features)
tsne_bow_features_df.columns = ["C1", "C2"]
tsne_bow_features_df["TITLE"] = bow_features_df["TITLE"]
tsne_bow_features_df["ID"] = bow_features_df["ID"]

display(tsne_bow_features_df)
```

	C1	C2	TITLE	ID
0	4.678334	0.676908	Ensemble methods: bagging, boosting and stacking	1
1	-0.525218	-0.337440	Understanding AUC - ROC Curve	2
2	3.179588	0.410891	How to work with object detection datasets in ...	3
3	4.088364	-0.551824	11 Dimensionality reduction techniques you sho...	4
4	1.404219	-1.041643	The Time Series Transformer	5
...	...	...	...	...
203	-5.789235	-2.580537	Type 2 Diabetes Reversal The Quick Start Guide	210
204	-5.116230	-1.458020	How a 22 Day Water Fast Changed My Life	211
205	-4.967505	-1.612653	Breaking Your Fast	212
206	-4.958772	0.348647	11 Unusual Tips for How to Wake Up Early	213
207	-6.518097	-1.696942	The 3 Biggest Mistakes Women Make On The Ketog...	214

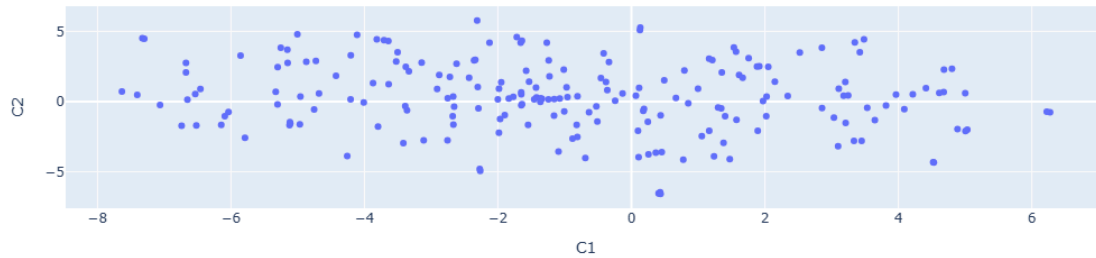
208 rows × 4 columns

```
In [35]: # scatter plot of t-sne for the BOW representation
import plotly.express as px

title = "T-distributed Stochastic Neighbor Embedding for BOW document representatio

fig = px.scatter(tsne_bow_features_df, x="C1", y="C2", hover_data=['TITLE'], title=
fig.show()
```

T-distributed Stochastic Neighbor Embedding for BOW document representation



From the above graph, We see most of documents to the one half describe articles related to Deep learning. Articles plotted to the other half describe Coronavirus, Economics and other miscellaneous articles.

Similarity between documents using BOW representation.

```
In [36]: from sklearn.metrics.pairwise import cosine_similarity

def get_similar_documents(all_article_rep_df, article_id, features):
    # extracting features of a article
    this_article_rep = all_article_rep_df[all_article_rep_df["ID"] == article_id][f
    other_article_rep = all_article_rep_df[all_article_rep_df["ID"] != article_id][f
    # calculating cosine similarity
    similarity_matrix = cosine_similarity(this_article_rep, other_article_rep)
    similar_articles = list(zip(similarity_matrix[0].tolist(), all_article_rep_df["
    # sorting
    similar_articles = sorted(similar_articles, key = lambda x : x[0], reverse = Tr
    print("Reference Article : {}".format(all_article_rep_df[all_article_rep_df["ID

    print("**** Similar Articles ****")
    # top 5 similar articles
    for score, title in similar_articles[:5]:
        print(title)
    print()

    # Let us check top 5 similar articles for some of the articles in our corpus
    get_similar_documents(bow_features_df, 90, count_vectorizer.get_feature_names_out()
    get_similar_documents(bow_features_df, 80, count_vectorizer.get_feature_names_out()
    get_similar_documents(bow_features_df, 150, count_vectorizer.get_feature_names_out(
    get_similar_documents(bow_features_df, 205, count_vectorizer.get_feature_names_out(
```

Reference Article : 17 types of similarity and dissimilarity measures used in data science.

**** Similar Articles ****

9 Distance Measures in Data Science

17 Clustering Algorithms Used In Data Science and Mining

Machine Learning Basics with the K-Nearest Neighbors Algorithm

OVER 100 Data Scientist Interview Questions and Answers!

Fundamental Techniques of Feature Engineering for Machine Learning

Reference Article : TensorFlow Tutorial Part 1

**** Similar Articles ****

How to go from a Python newbie to a Google Certified TensorFlow Developer under two months

Enchanted Random Forest

The 7 Best Data Science and Machine Learning Podcasts

PCA using Python (scikit-learn)

Time Series Forecasting with PyCaret Regression Module

Reference Article : The One Word That Explains Why Economics Professors Are Not Billionaires

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Why People Still Don't Get Cryptocurrency

You May Have A Poor Person's Mindset And Not Know It

You Will Never Be Rich If You Keep Doing These 10 things

How I transformed \$6,000 to \$3,000,000

The Exact Steps I Followed to Make \$1,500+ of Passive Income Every Month

Reference Article : How I lost 10kg in 60 days: My 7-step weight loss plan

**** Similar Articles ****

10 Things You Can Do This Morning To Heal Your Anxiety

How to lose 10+ pounds of fat a month- even if you have a slow metabolism

The Diet & Workout Plan of a Full-Time Traveling Family HIS

The cure for type 2 diabetes is known, but few are aware

The Easiest Way to Lose 125 Pounds Is to Gain 175 Pounds

The suggested articles do seem to be similar to those in reference. BOW looks for presence and absence of words in different documents, so documents which have similar words would also have similar BOW representation, and hence would be more similar.

In []:

Similarity using TF-IDF

```
In [38]: # TF-IDF representation of the dataset Using TfidfVectorizer from scikit-Learn
from sklearn.feature_extraction.text import TfidfVectorizer
tfidf_vectorizer = TfidfVectorizer(min_df=5)
# min_df: ignore terms that have a document frequency strictly lower than the given

# todense() returns a matrix
tfidf_features = tfidf_vectorizer.fit_transform(articles["processed_text"]).todense()

# create dataframe
tfidf_features_df = pd.DataFrame(tfidf_features)
tfidf_features_df.columns = tfidf_vectorizer.get_feature_names_out() # Get output
tfidf_features_df["TITLE"] = articles["title"]
```

```
tfidf_features_df["ID"] = articles["id"]  
display(tfidf_features_df)
```

	00	000	01	05	06	07	10	100	1000	101	10k	10x	11
0	0.000000	0.0	0.0	0.0	0.000000	0.0	0.000000	0.0	0.0	0.0	0.0	0.0	0.000000
1	0.000000	0.0	0.0	0.0	0.000000	0.0	0.000000	0.0	0.0	0.0	0.0	0.0	0.000000
2	0.000000	0.0	0.0	0.0	0.019284	0.0	0.008320	0.0	0.0	0.0	0.0	0.0	0.000000
3	0.000000	0.0	0.0	0.0	0.000000	0.0	0.000000	0.0	0.0	0.0	0.0	0.0	0.006737
4	0.000000	0.0	0.0	0.0	0.000000	0.0	0.000000	0.0	0.0	0.0	0.0	0.0	0.000000
...	...	...	...	...	...	...	...	...	...	...	...	...	...
203	0.015154	0.0	0.0	0.0	0.000000	0.0	0.006358	0.0	0.0	0.0	0.0	0.0	0.000000
204	0.000000	0.0	0.0	0.0	0.000000	0.0	0.000000	0.0	0.0	0.0	0.0	0.0	0.000000
205	0.000000	0.0	0.0	0.0	0.000000	0.0	0.016279	0.0	0.0	0.0	0.0	0.0	0.022684
206	0.000000	0.0	0.0	0.0	0.000000	0.0	0.043608	0.0	0.0	0.0	0.0	0.0	0.000000
207	0.000000	0.0	0.0	0.0	0.000000	0.0	0.007903	0.0	0.0	0.0	0.0	0.0	0.000000
208 rows × 3715 columns													

```
In [39]: tsne = TSNE(n_components=2)
tsne_tfidf_features = tsne.fit_transform(tfidf_features_df[tfidf_vectorizer.get_fea
tsne_tfidf_features_df = pd.DataFrame(tsne_tfidf_features)
tsne_tfidf_features_df.columns = ["C1", "C2"]
tsne_tfidf_features_df["TITLE"] = tfidf_features_df["TITLE"]
display(tsne_tfidf_features_df)
```

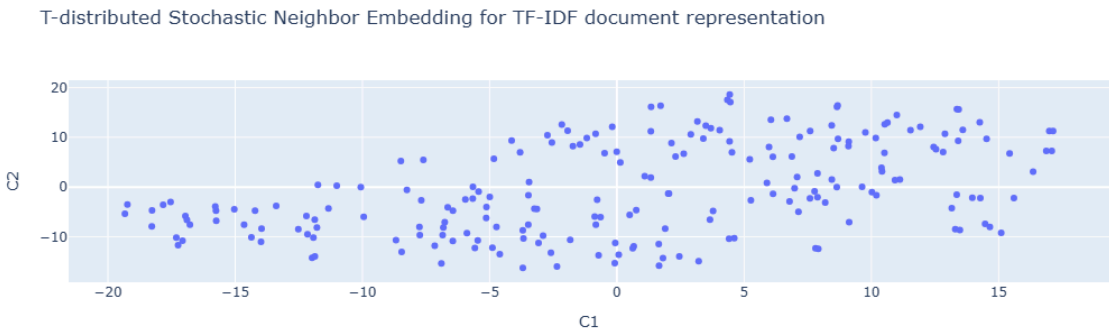
	C1	C2	TITLE
0	9.762344	11.015292	Ensemble methods: bagging, boosting and stacking
1	14.255919	13.017860	Understanding AUC - ROC Curve
2	6.873959	6.165265	How to work with object detection datasets in ...
3	12.444081	8.057103	11 Dimensionality reduction techniques you sho...
4	3.163160	13.177513	The Time Series Transformer
...	...	...	...
203	-19.331902	-5.387913	Type 2 Diabetes Reversal The Quick Start Guide
204	-15.034296	-4.478994	How a 22 Day Water Fast Changed My Life
205	-15.781659	-3.918873	Breaking Your Fast
206	-12.157654	-9.466602	11 Unusual Tips for How to Wake Up Early
207	-17.537691	-3.023856	The 3 Biggest Mistakes Women Make On The Ketog...

208 rows × 3 columns

```
In [ ]:
```

Viualization for similarity using TF-IDF

```
In [41]: title = "T-distributed Stochastic Neighbor Embedding for TF-IDF document representa
fig = px.scatter(tsne_tfidf_features_df, x="C1", y="C2", hover_data=['TITLE'], titl
fig.show()
```



From the above scatter plot, we see points usually along three different directions. while hovering over the points,we see the three directions contains articles broadly on the following topics - Data Science/Machine Learning Health Finance

In [43]: *# Let us check top 5 similar articles for some of the articles in our corpus*

```
get_similar_documents(tfidf_features_df, 90, tfidf_vectorizer.get_feature_names_out
get_similar_documents(tfidf_features_df, 80, tfidf_vectorizer.get_feature_names_out
get_similar_documents(tfidf_features_df, 150, tfidf_vectorizer.get_feature_names_ou
get_similar_documents(tfidf_features_df, 205, tfidf_vectorizer.get_feature_names_ou
```

Reference Article : 17 types of similarity and dissimilarity measures used in data science.

**** Similar Articles ****

9 Distance Measures in Data Science

17 Clustering Algorithms Used In Data Science and Mining

Machine Learning Basics with the K-Nearest Neighbors Algorithm

OVER 100 Data Scientist Interview Questions and Answers!

Fundamental Techniques of Feature Engineering for Machine Learning

Reference Article : TensorFlow Tutorial Part 1

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How to go from a Python newbie to a Google Certified TensorFlow Developer under two months

Enchanted Random Forest

Building RNN, LSTM, and GRU for time series using PyTorch

PCA using Python (scikit-learn)

Yes you should understand backprop

Reference Article : The One Word That Explains Why Economics Professors Are Not Billionaires

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Why People Still Don't Get Cryptocurrency

You May Have A Poor Person's Mindset And Not Know It

How I transformed \$6,000 to \$3,000,000

4 Reasons Why Economists Make Great Data Scientists (And Why No One Tells Them)

Reference Article : How I lost 10kg in 60 days: My 7-step weight loss plan

**** Similar Articles ****

How to lose 10+ pounds of fat a month- even if you have a slow metabolism

10 Things You Can Do This Morning To Heal Your Anxiety

The Diet & Workout Plan of a Full-Time Traveling Family HIS

The Science Behind Fat Metabolism

The cure for type 2 diabetes is known, but few are aware

TFIDF also seems to work well.

- Noticed one improvement in the articles suggested for "The One Word That Explains Why Economics Professors Are Not Billionaires"
- Also noticed the improvement in recommendations suggested for "TensorFlow Tutorial Part 1"
- All 5 articles suggested using TF-IDF seems more relevant as compared to articles suggested by BOW.

Conclusion: Medium articles in our corpus mainly belonged to Data Science, Health and Finance. In other words there are three (or maybe more) broad topics (Data Science/Machine Learning, Health and Finance), that our corpus talks about - this can be

segregated or identified using Topic Modeling. The vectorization techniques that we talked about are used extensively in text classification problems as well.

In []:

In []:

In []:

In []:

In []:

In []: